

How gamma controls access to representable target components

The proposed paper claim. A target component can belong to a frozen tanh dictionary's span and still require millions of readout updates to learn. A common small slope γ suppresses frequency content in the features. On a finite interval, sampling and boundaries mix those frequencies; retaining this mixing carries the explicit attenuation into accurate eigenvalues and target projections. We can therefore identify **representable but slow target components**, and predict the resulting delay before running GD.

For a paper with room for one theorem, we recommend the **finite-interval tanh theorem** below. It concerns the features actually trained. The periodic calculation is a motivating example and a step in the proof, rather than a second headline theorem. Evaluating the result includes a finite spectral calculation. No decay rate is fitted to the observed training curve.

The figure follows three steps: gamma filters frequency content; the finite kernel places target energy in positive but slow modes; that spectrum predicts the observed training trajectory. At gamma 8, the calculation places at least **4.3322% of target energy in representable slow modes**, implying a **3.036-million-update necessary time** from that band alone. The complete spectrum gives a narrow interval containing the measured **15,798,313 updates**. Spectral numbers use a disclosed floating-point allowance; the four primary full-spectrum timing intervals also contain independently certified archived intervals.

Notation. J_γ is the sample-normalized feature matrix; $K_\gamma = J_\gamma J_\gamma^*$ is its readout kernel. μ_j is an eigenvalue, and $\eta\mu_j$ is its per-update learning rate. $E_\gamma(n)$ is relative residual norm: 1% means $E_\gamma(n) \leq 0.01$. Physical slope is γ ; dimensionless bandwidth is $\lambda = \gamma h$, never an eigenvalue. A hat denotes the constructed approximation; δ_γ bounds its kernel error. The band-energy fraction is p_γ ; the periodic reference feature is ϕ_γ .

1 Why the periodic example makes the mechanism transparent

A tanh is exactly a smoothed step:

$$\tanh(\gamma x) = (\rho_\gamma * \text{sign})(x), \quad \rho_\gamma(t) = \frac{\gamma}{2} \text{sech}^2(\gamma t).$$

The smoothing density has Fourier multiplier

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1.$$

On a circle, with centers and input measure uniform over the whole period, translations make the kernel diagonal in Fourier waves. Smoothing multiplies each feature coefficient by M_γ , and the kernel pairs two coefficients:

$$\mu_\omega(\gamma) = \mu_\omega^{\text{step}} M_\gamma(\omega)^2.$$

At high frequency relative to gamma,

$$M_\gamma(\omega)^2 \sim \left(\frac{\pi|\omega|}{\gamma} \right)^2 e^{-\pi|\omega|/\gamma}.$$

At a fixed step size, this produces an exponential optimization penalty. Every frequency with $\mu_\omega^{\text{step}} > 0$ remains accessible at every $\gamma > 0$, but its learning rate can become extremely small. Frequencies absent from the periodic step remain inaccessible; for the standard square wave these include nonconstant even harmonics. Bias supplies the constant mode. The claim concerns accessible frequencies, not all periodic functions.

Ordinary tanh on an interval is not this periodic population model. Boundaries and finite sampling can rotate learning directions. The finite theorem retains these effects instead of treating periodic eigenvalues as an approximation to the answer.

2 The one theorem to present

Fix an interval of length $L > 0$, an integer $N \geq 3$, and equally spaced core centers $c_j = jh$, $j = 0, \dots, N-1$, $h = L/N$, and inputs $x_i = ih/q$, where q is a positive integer and $i = 0, \dots, qN$. Thus $m = qN + 1$ includes both endpoints. Any extra centers stay fixed and are retained explicitly; W counts all tanh features. Include a bias column equal to $1/\sqrt{m}$ and define

$$J_\gamma[i, j] = \frac{\tanh(\gamma(x_i - c_j))}{\sqrt{m}}, \quad y_i = \frac{y_i^{\text{phys}}}{\sqrt{m}}.$$

All tanh features have a common frozen slope $\gamma > 0$. GD minimizes $\frac{1}{2}\|J_\gamma w - y\|^2$ in the original Euclidean readout coefficients, starts at $w = 0$, and uses $\eta > 0$. The nonzero target y and all input/center locations stay fixed as gamma changes.

Its relative residual is $E_\gamma(n) = \|(I - \eta K_\gamma)^n y\|/\|y\|$. Write μ_j for eigenvalues in decreasing order, including zeros, and u_j for an orthonormal eigenbasis. Define cumulative target energy

$$S_\gamma(t) = \sum_{\eta\mu_j \leq t} \frac{|\langle y, u_j \rangle|^2}{\|y\|^2}.$$

First synthesize a periodic reference from Fourier coefficients multiplied by M_γ . Then retain the gamma-dependent boundary terms, exact boundary columns, extra centers, bias, and endpoint row. Call the resulting features $\widehat{J}_\gamma$ and kernel $\widehat{K}_\gamma$. Appendix A gives the scalar formulas and error allowance; no training observation is used.

Theorem 1 (Gamma-dependent access to representable finite-tanh components). Under the stated geometry and raw-readout parameterization, the Fourier and boundary construction gives an explicit allowance δ_γ such that

$$\|K_\gamma - \widehat{K}_\gamma\| \leq \delta_\gamma, \quad |\mu_j(\gamma) - \widehat{\mu}_j(\gamma)| \leq \delta_\gamma.$$

All gamma dependence is explicit in the frequency multipliers, boundary factors, and allowance. The analytic allowance decreases geometrically to zero as the retained boundary expansion increases.

Choose $\eta(\|\widehat{K}_\gamma\| + \delta_\gamma) \leq 1$. Let $\underline{S}_\gamma(t)$ and $\overline{S}_\gamma(t)$ be the bounds defined below on the fraction of target energy in per-update rates at most t . For $0 < a < b < 1$, set

$$p_\gamma = \left[\underline{S}_\gamma(b) - \overline{S}_\gamma(a) \right]_+.$$

At least a fraction p_γ of target energy lies in the positive-rate band $(a, b]$. This component is exactly representable on the sampled grid. Nevertheless, it implies

$$\boxed{E_\gamma(n) \geq \sqrt{p_\gamma}(1 - b)^n}.$$

The theorem identifies a property of the specified target and dictionary. It does not assume target alignment with a chosen Fourier wave. Gamma changes both eigenvalues and finite eigenvectors; the spectral calculation retains both changes when computing target energy.

For explicit energy bounds, compute the eigenvectors $\widehat{u}_j$ of $\widehat{K}_\gamma$ and set

$$\widehat{S}_\gamma(t) = \sum_{\eta\widehat{\mu}_j \leq t} \frac{|\langle y, \widehat{u}_j \rangle|^2}{\|y\|^2}.$$

Include the full nullspace. For a chosen buffer $g > 0$, valid bounds are

$$\begin{aligned} \underline{S}_\gamma(t) &= \left(\sqrt{\widehat{S}_\gamma(t - g)} - \frac{\eta\delta_\gamma}{g} \right)_+^2, \\ \overline{S}_\gamma(t) &= \min \left\{ 1, \left(\sqrt{\widehat{S}_\gamma(t + g)} + \frac{\eta\delta_\gamma}{g} \right)^2 \right\}. \end{aligned}$$

Buffers may differ between cutoffs. Taking the largest lower and smallest upper bound over any declared collection remains valid. Subtracting upper mass at a strictly positive a is essential: it excludes the possibility that the reported slow energy is merely an unlearnable nullspace component.

The full target need not be exactly representable. The component isolated by $(a, b]$ is. This precisely separates optimization access from capacity.

3 Where gamma enters, and what must still be calculated

The quantitative chain is

$$\begin{aligned} \gamma &\longrightarrow \underbrace{M_\gamma(\omega) \text{ and boundary factors}}_{\text{explicit feature construction}} \\ &\longrightarrow \underbrace{\widehat{\mu}_j(\gamma), \widehat{u}_j(\gamma)}_{\text{finite spectral calculation}} \longrightarrow \underbrace{p_\gamma \text{ and } E_\gamma(n)}_{\text{target access and delay}} . \end{aligned}$$

This goes beyond observing that gamma changes a matrix. We identify its feature-level action analytically, retain the geometry that mixes frequencies, and bound the resulting spectral error. Finite eigenvectors are computed, not assumed to remain Fourier waves. The boundary correction also depends on gamma; it is neither a free fit nor an assumed small perturbation.

Two errors must not be confused. Small gamma can make true learning rates small. Separately, truncation introduces approximation error δ_γ , reducible by retaining more terms. The latter controls prediction precision; it is not the proposed cause of delay.

The periodic example gives a scalar cap-to-rate inequality immediately. On the finite interval, the theorem gives quantitative predictions after solving the explicit corrected spectral problem. It does not assert that every target or finite eigenvalue improves monotonically with gamma. Such ordering is unnecessary to explain the measured, target-dependent delay.

4 Predictions and the three-panel figure

The primary target is

$$f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x), \quad x \in [-1, 1].$$

The archive uses $L = 2$, $N = 512$, $q = 16$, and $m = 8193$. Its 512 core centers, terminal center, and 23 halo centers on each side give 559 tanh features plus bias. Common frozen slopes are 8, 12, 16, and 64. Readouts start at zero; each saved step is curvature-normalized to approximately $\eta \|K_\gamma\| = 0.5$. Comparisons therefore concern those actual normalized steps, not one common raw step. Translating $[-1, 1]$ to $[0, L]$ changes neither feature differences nor target samples.

At $\gamma = 8$, $a = 10^{-9}$, $b = 10^{-6}$, numerical evaluation gives

$$\underline{S}_8(b) = 0.0433317719, \quad \overline{S}_8(a) = 0.00000929944, \quad p_8 \geq 0.0433224724.$$

Thus at least 4.3322% of target energy lies in strictly positive but slow modes. This band requires at least **3,035,627 updates** to reach 1% relative residual. At gamma 64, even the upper bound on all energy with rates at most 10^{-6} is only about **0.000162%**. These compare the same target, retaining each gamma's own finite directions.

The single-band bound discards variation within that band. Keeping every rate and target weight gives the necessary–sufficient intervals

$$\begin{aligned} [15,767,081, 15,829,763], & \quad [186,056, 186,059], \\ [61,792, 61,793], & \quad [16,013, 16,013]. \end{aligned}$$

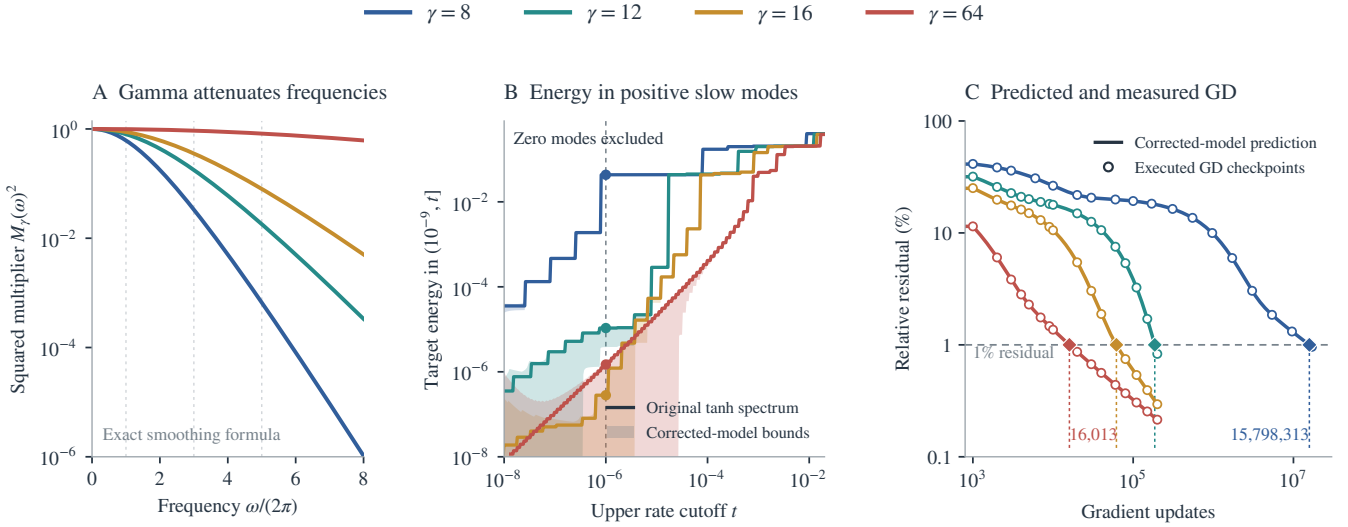


Figure 1: Gamma changes optimization access to a fixed target. A shows exact analytic periodic attenuation, not measured training data. Faint guides mark the target’s physical frequencies, without identifying them with finite-kernel eigenvectors. B shows finite ordinary-tanh target energy in positive rates above $a = 10^{-9}$ and below the horizontal-axis cutoff. Solid curves come from the original tanh spectrum; shaded bounds come from the Fourier-plus-boundary construction. The marked cutoff is $b = 10^{-6}$. C compares predictions from the complete constructed spectrum with saved GD checkpoints. The horizontal 1% threshold and crossing markers show acquisition time. Circles are saved GD measurements, diamonds are executed first hits, and short horizontal segments at 1% are necessary–sufficient timing intervals. Colors identify the same four slopes throughout. Spectral bands use the disclosed FP64 sensitivity allowance, not independent interval certification. The primary full-spectrum timing intervals contain previously certified intervals.

Measured crossings are **15,798,313**, **186,057**, **61,792**, and **16,013**. The gamma-8 interval width is 0.397% of its observed time. It is useful to show both the positive slow component explaining the obstruction and the full forecast predicting its duration.

This retrospectively analyzes existing optimization experiments. The representative target and rate band were chosen during analysis, not preregistered. Truncations use analytic error bounds; the buffer scan uses the constructed spectrum without consulting the original target mass. There is no new training or fitted decay parameter, and no held-out generalization claim. The wider package has three grid sizes, four slopes, and five targets, with all 60 reference crossings and all 19 available executed crossings enclosed.

The spectral bands add a heuristic FP64 arithmetic allowance to the proved analytic remainder; they are not independent interval certificates. The four primary full-spectrum timing intervals also contain independently certified Arb intervals, validating the wider timing intervals by monotonicity. This inherited validation does not certify the new positive-band energy numbers. The [full technical note](#) records the arithmetic allowance and its tenfold sensitivity check.

A Explicit construction and proof

The reference and its boundaries. Let s be the $2L$ -periodic square wave equal to $\text{sign}(d)$ on $(-L, L)$, and $\phi_\gamma = \rho_\gamma * s$. It is antiperiodic: shifting by L flips its sign. Its Fourier expansion is

$$\phi_\gamma(d) = \sum_{k \in \mathbb{Z}} \frac{2}{i\pi(2k+1)} M_\gamma\left(\frac{(2k+1)\pi}{L}\right) e^{i(2k+1)\pi d/L}.$$

Positive gamma gives absolute convergence. Sampling retains aliases: coefficients of frequencies indistinguishable on the finite grid add before forming the kernel. Do not substitute population eigenvalues for this sampled calculation.

The exact image identity is

$$\phi_\gamma(d) = \tanh(\gamma d) + \sum_{\ell \geq 1} (-1)^\ell \left[\tanh(\gamma(d - \ell L)) + \tanh(\gamma(d + \ell L)) \right].$$

Both sides have derivative $\sum_{\ell \in \mathbb{Z}} (-1)^\ell \gamma \operatorname{sech}^2(\gamma(d - \ell L))$. Both vanish at zero and their paired tails converge locally uniformly, proving the identity. Expanding each image's positive-distance tanh as a geometric series gives, for $|d| < L$,

$$\tanh(\gamma d) = \phi_\gamma(d) + 2 \sum_{r \geq 1} \frac{(-1)^{r-1}}{1 + e^{-2r\gamma L}} \left[e^{-2r\gamma(L-d)} - e^{-2r\gamma(L+d)} \right].$$

This is explicit finite-interval gamma dependence: the Fourier attenuation and every boundary coefficient are known functions of gamma. With $d = x - c$, each exponential separates into a factor of x and one of c . Retaining p terms gives correction rank at most $2p$ on interior columns.

A controllable remainder. Choose an integer B with $1 \leq B < N/2$ and a nonnegative integer p . Treat the first and last B core columns exactly. On the other core columns and all but the endpoint input row, use ϕ_γ plus the first p boundary terms. Treat extra centers, bias, and the endpoint row exactly. This defines $\widehat{J}_\gamma$ in the original feature coordinates.

Interior distances have $|d| \leq L - Bh$. The finite geometric remainder of each tanh expansion is at most twice its first omitted exponential. Summing both sides and all images gives the unnormalized entry-error bound

$$\epsilon_{B,p}(\gamma) = \frac{4e^{-2(p+1)\gamma Bh}}{1 - e^{-2(p+1)\gamma L}}.$$

At most $N - 2B$ columns have error. Frobenius norm bounds operator norm, so

$$\epsilon_J = \sqrt{N - 2B} \epsilon_{B,p}(\gamma), \quad \|J_\gamma - \widehat{J}_\gamma\| \leq \epsilon_J.$$

Every true unnormalized entry has magnitude at most one, including bias. Thus $\|J_\gamma\| \leq \sqrt{W + 1}$ and the expanded Gram difference gives

$$\boxed{\delta_\gamma = 2\sqrt{W + 1} \epsilon_J + \epsilon_J^2.}$$

This proves the kernel bound; the min-max characterization proves the ordered eigenvalue bound. If the reference Fourier series is truncated with uniform entry error τ , replace $\epsilon_{B,p}$ by $\epsilon_{B,p} + \tau$ before forming ϵ_J and δ_γ . Floating-point evaluation needs a separate allowance.

The reference's uniform-grid blocks are explicitly diagonalizable by half-odd Fourier vectors. Restoring boundary columns, the p separable terms, $W - N$ extra centers, bias, and endpoint row changes the feature matrix by rank at most $2B + 2p + (W - N) + 2$. The kernel correction rank is at most twice this. It is signed and need not be small in norm. The [full construction proof](#) supplies exact alias sums and signed spectral formulas. The numerical full forecast uses the SVD of the constructed feature matrix; it does not claim to obtain all target projections from a small eigenvalue-count calculation.

Transferring target energy. Put $A = \eta K_\gamma$, $\widehat{A} = \eta \widehat{K}_\gamma$, and $d = \eta \delta_\gamma$. Let P project onto eigenvalues of A at most t , and Q onto eigenvalues of $\widehat{A}$ at most $t - g$. The cross-projection $X = (I - P)Q$ solves

$$A_{>t} X - X \widehat{A}_{\leq t-g} = (I - P)(A - \widehat{A})Q.$$

The restricted spectra are separated by at least g . The exponential integral solution of this Sylvester equation gives $\|X\| \leq d/g$. Therefore $\|Qy\| \leq \|Py\| + (d/g)\|y\|$, yielding the lower cumulative bound. Use the projector of $\widehat{A}$ below $t + g$ and reverse the roles for the upper bound. This controls subspaces without requiring a gap between every pair of eigenvalues.

True energy in $(a, b]$ is $S_\gamma(b) - S_\gamma(a)$ and is at least $[\underline{S}_\gamma(b) - \bar{S}_\gamma(a)]_+$. Every eigenvalue in this band is positive, so its eigenvector belongs to $\operatorname{range}(K_\gamma) = \operatorname{range}(J_\gamma)$. This proves exact representability of the isolated component on the input grid.

Transferring to training. Zero initialization gives $r_n = (I - \eta K_\gamma)^n y$. Under the step condition, each rate in $(a, b]$ retains at least a factor $1 - b$ of residual amplitude per update. Square and sum the target coefficients to obtain $E_\gamma(n)^2 \geq p_\gamma (1 - b)^{2n}$. For $p_\gamma > \varepsilon^2$, reaching $E_\gamma(n) \leq \varepsilon$ requires

$$n \geq \left\lceil \frac{\log(\sqrt{p_\gamma}/\varepsilon)}{-\log(1 - b)} \right\rceil.$$

The complete constructed spectrum gives

$$\widehat{E}_\gamma(n)^2 = \sum_j \frac{|\langle y, \widehat{u}_j \rangle|^2}{\|y\|^2} (1 - \eta \widehat{\mu}_j)^{2n},$$

$$|E_\gamma(n) - \widehat{E}_\gamma(n)| \leq n\eta\delta_\gamma.$$

The sum includes nullspace energy. Telescope the two matrix powers: each of n terms contains one kernel difference and otherwise contractions. This proves the last inequality. Thus $\widehat{E}_\gamma(n) + n\eta\delta_\gamma \leq \varepsilon$ proves acquisition by n , while $\widehat{E}_\gamma(n) - n\eta\delta_\gamma > \varepsilon$ excludes it by n . This is the origin of the necessary–sufficient timing intervals.

B Paper placement and reproducibility

Use the periodic multiplier as a motivating example, the finite-tanh access result as the sole numbered theorem, and the three-panel figure as its empirical evaluation. Keep the scalar construction, transfer proof, full timing formula, and numerical sensitivity in the appendix. Present common frozen slope, uniform core centers and samples, fixed target, and raw readout GD alongside the theorem. Representability means representability on the training grid.

At grid-relative frequency $\theta = \omega h$, attenuation depends on γh through $z = \pi|\theta|/(2\gamma h)$. Holding γh fixed preserves this part of spectral access under refinement. It does not alone remove raw-kernel width scaling, boundary effects, or step dependence.

The [figure data](#) records source hashes, positive-band bounds, GD checkpoints, and the distinction between analytic and measured quantities. Reproduce the figure with

```
python -m experiments.expD36_frozen_gamma_probe.gamma_access_figure
```

Compile this note with

```
latexmk -pdf -outdir=/tmp/gamma-access-latex
docs/gamma_access_collaborator_note.tex
```

Earlier proof notes and archives remain unchanged. No new GPU training is used.